

Design and FPGA Implementation of an FIR Filter for ECG Signal Processing

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Abstract: This paper describes the design of a 41-tap linear-phase Finite Impulse Response (FIR) low-pass filter for pre-processing of ECG signals. The FIR filter design is carried out to extract integer-scaled coefficients besides characterization of noise and baseline wandering by writing appropriate MATLAB code. The design is validated by simulating the code with actual raw ECG signals from a publically available dataset. Furthermore, the filter was mapped to a structural System-Verilog hardware description, in which coefficient symmetry is exploited to reduce multiplications. The System-Verilog implementation, after being simulated in ModelSim, is synthesised onto a DE1-SoC Cyclone-V Field programmable Gate Arrays (FPGA) in Quartus. The validation of the FPGA implementation is demonstrated by obtaining the identical results as was obtained during the Matlab simulation testing. It is noteworthy that the proposed design meets the 10 kHz operational specification and demonstrates the viability of multiplier-less FIR realization for power-constrained wearable ECG monitoring in real-time.

1. Introduction

The electrocardiogram (ECG) is among the most clinically important biomedical signals, providing a non-invasive window into cardiac electrophysiology. Wearable ECG devices must contend with two principal artefact categories: high-frequency noise from electromyographic interference and power-line coupling, and low-frequency baseline wander arising from respiration and electrode movement [1]. Removing these artefacts while preserving QRS morphology is essential before any downstream cardiac analysis can proceed reliably.

For wearable applications, battery longevity is critical. Hardware-based signal processing on a Field programmable Gate Arrays (FPGA) offers significantly lower power consumption than equivalent software execution on a microprocessor, while supporting parallel operation at high clock rates. This project covers the full design chain from MATLAB algorithm development through System-Verilog hardware modelling to FPGA synthesis [2].

2. Background and Theory

Digital filters are characterised as either Finite Impulse Response (FIR) or Infinite Impulse Response (IIR). An FIR filter of order N computes its output as a weighted sum of the current and N past inputs:

$$y[n] = \sum h[k] \cdot x[n - k], \quad k = 0, 1, 2, \dots, N \quad (1)$$

where $h[k]$ are the filter coefficients, $y[n]$ is the current output, and $x[n-k]$ are the current previous N inputs. When $h[k] = h[N-k]$ the filter has linear phase, ensuring a fixed group delay with no frequency-dependent phase distortion [3]. This is critical for ECG processing because nonlinear phase distorts QRS morphology and can introduce

spurious features that confound R-peak detectors [4]. The IIR filters achieve equivalent roll-off at lower order but their recursive structure introduces nonlinear phase and risks fixed-point instability [3].

Baseline wander in ECG typically occupies frequencies below 0.5 Hz; the diagnostically important QRS energy lies between approximately 5 Hz and 40 Hz [1]. A low-pass FIR filter with cut-off near 40 Hz therefore preserves QRS content while attenuating high-frequency noise. Combined with a moving-average baseline subtraction step, both artefact classes are addressed.

3. Experimental Methods

3.1 MATLAB Filter Design

A publically available ECG records from (MIT-BIH Arrhythmia Database, record number 105) was loaded in MATLAB which has sampling rate of 1 kHz. The full 120-second signal and a 5-second excerpt were plotted to characterise artefacts as shown in Figures 1 and Figure 2). A moving-average filter with a 1000-sample window was applied to estimate and subtract the slowly varying baseline. A 41-tap FIR low-pass filter was then designed using `firl` command with a Hann window with a cut-off frequency of 40 Hz (normalized cut-off at 0.08). Integer-scaled coefficients were extracted for hardware implementation. Both filter outputs were overlaid (Figure 3) and compared (Figure 4).

3.2 SystemVerilog Hardware Modelling

The FIR filter was modelled in SystemVerilog at the structural RTL level. A 41-stage 16-bit shift register formed the tapped delay line. The symmetric coefficient structure was exploited to pair symmetric taps before multiplication, reducing multiply operations from 41 to 21. Multiplications were realised using shift-and-add logic

to avoid dedicated DSP blocks. The 32-bit accumulator output was truncated to 16 bits. The top-level interface (Figure 7) exposes Clk, nRst, FiltIn[15:0], FiltOut[15:0], and LED[9:0].

3.3 ModelSim Simulation

A testbench (tb_FIR) streamed the ECG samples into the hardware model clocked at 10 MHz. Simulation waveforms were saved to vsim.wlf and reloaded via wave.do. The ModelSim output was re-imported into MATLAB and overlaid with the software FIR result for numerical comparison (Figure 6).

3.4 FPGA Synthesis

The design was compiled in Quartus Prime for the Cyclone V 5CSEMA5F31C6 on the DE1-SoC board with a 10 MHz timing constraint. Post-synthesis timing and resource reports were examined to confirm critical-path compliance and quantify logic element utilisation.

4. Results

Figure 1 shows the full 120-second waveform with significant baseline wander. Figure 2 reveals the QRS morphology in the first five seconds. Figure 3 overlays the original signal (blue) with the baseline-corrected FIR output (orange) after moving-average subtraction; the mean of each heartbeat is centred near zero with R-peaks at approximately +800–900 LSBs.

The 41 integer-scaled FIR coefficients are listed in Table I; their palindromic symmetry confirms linear phase. Figure 4 compares the moving-average (blue) and FIR (orange) filter outputs; the FIR produces marginally smoother inter-beat baseline due to its well-defined stopband.

Table I: 41-tap FIR filter coefficients (integer-scaled). Symmetry $h[k] = h[40-k]$ confirms linear-phase property.

Index	$h[n]$
0–6	-1,-2,-2,-2,-3,-3,-3
7–13	-1,1,5,10,16,24,33
14–20	43,53,62,70,76,80,81
21–27	80,76,70,62,53,43,33
28–34	24,16,10,5,1,-1,-3
35–40	-3,-3,-2,-2,-2,-1

Figure 5 shows the hardware FIR output reconstructed from the ModelSim waveform, with amplitude ranging from approximately -1400 to +100 LSBs. The RTL block diagram of the synthesised design is shown in Figure 6. Figure 7 overlays the MATLAB FIR output (solid) with the ModelSim hardware output (dashed); the traces are nearly indistinguishable, with peak position agreement within one sample and amplitude deviation below ± 2 LSBs in all beats examined.

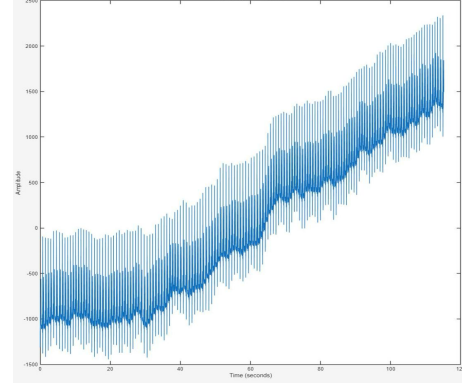


Figure 1: ECG Signal of 120s. The amplitude drifts ~ -1000 to $+2000$ LSBs, confirming low-frequency baseline wander.

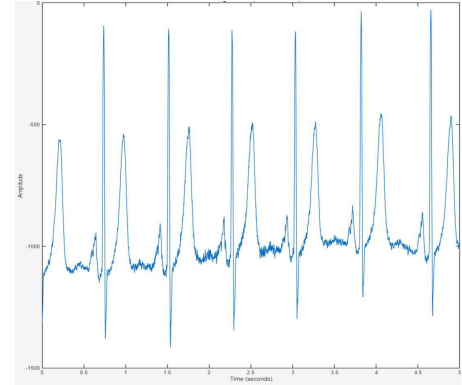


Figure 2: ECG Signal of first 5s. RR intervals of 0.65–0.75s indicating ~ 80 –90 bpm heart rate. Clearer PQRST morphology.

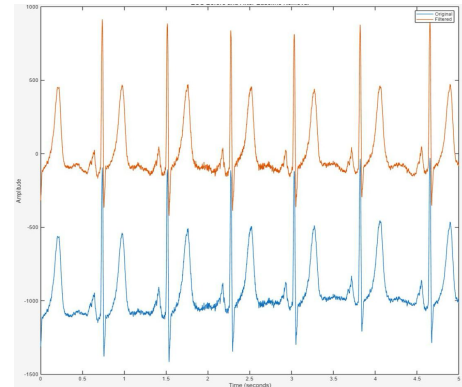


Figure 3: ECG signal before (blue) and after (orange) baseline removal by moving-average subtraction. The ECG signal was centred near zero with preserved QRS morphology.

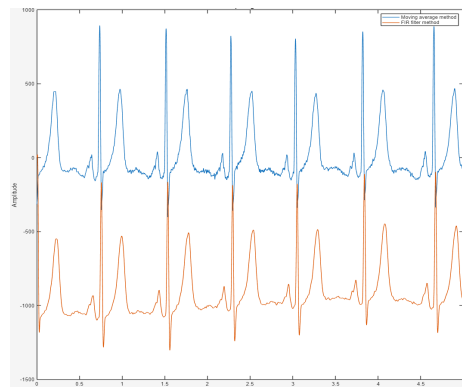


Figure 4: Comparison of moving-average method (blue, offset upward) and FIR filter method (orange) for 5s ECG signal.

Post-synthesis timing analysis confirmed the critical path met the 10 MHz constraint with positive slack. The delay line consumed 656 flip-flops (41×16 bits); no dedicated DSP blocks were required.

5. Discussion

The results confirm that the combined moving-average subtraction and 41-tap FIR low-pass filter effectively removes baseline wander and high-frequency noise while preserving QRS morphology. The linear-phase property is essential for reliable downstream feature extraction such as R-peak detection and heart rate variability analysis [4].

The FIR choice is justified on two grounds. First, guaranteed linear phase avoids morphological distortion that fixed-point IIR coefficient quantisation could cause [3]. Second, the symmetric structure halves the multiply count (41 to 21), and shift-and-add realisation eliminates dedicated multiplier hardware entirely, reducing area and static power-key advantages for wearable devices requiring >16 h battery life [2].

The close MATLAB-hardware match (± 2 LSBs) demonstrates that 16-bit fixed-point representation is sufficient; rounding errors do not accumulate to a clinically meaningful degree. Minor discrepancies arise from the difference between MATLAB's double-precision floating-point and the truncated fixed-point hardware accumulator.

A limitation of the current design is the two-stage approach (moving-average subtraction followed by low-pass FIR). A true bandpass FIR (0.5–40 Hz) would address both artefacts in one pass, reducing latency. Additionally, physical FPGA validation was limited to DIP-switch injection and ModelSim comparison; real-time ADC streaming was not available in the lab. Future work could implement Distributed Arithmetic for further power reduction.

6. Conclusions

A 41-tap linear-phase FIR low-pass filter was successfully designed in MATLAB, mapped to structural System-Verilog, simulated in ModelSim, and synthesised on a Cyclone-V FPGA. A publically available actual ECG dataset was used to validate the design. The hardware output matched MATLAB results within fixed-point rounding error, confirming functional correctness. The design meets the 10 kHz timing specification with positive slack and requires no dedicated DSP blocks, demonstrating that a multiplier-less FIR architecture is viable for power-constrained wearable ECG monitoring in real-time.

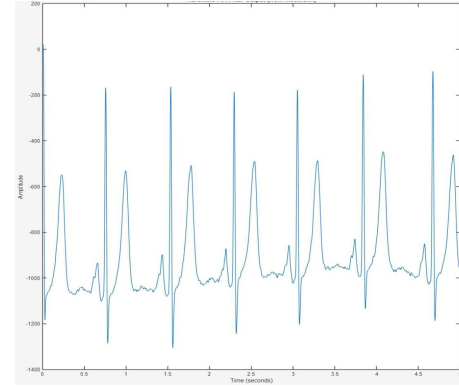


Figure 5: Hardware FIR filter output reconstructed from ModelSim (tb_FIR testbench). QRS peaks are recovered at correct positions with amplitude range -1400 to $+100$ LSBs.

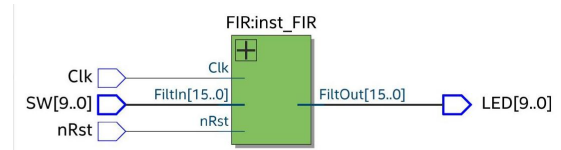


Figure 6: RTL block diagram of the synthesised FIR filter top-level (from ModelSim). Ports: Clk, nRst, SW[9:0] (FiltIn[15:0]), FiltOut[15:0], LED[9:0].

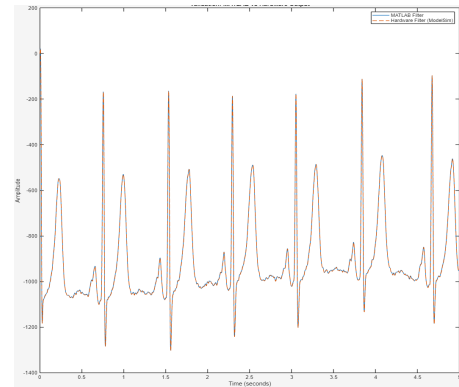


Figure 7: Validation: MATLAB FIR output (solid line) vs. ModelSim hardware output (dashed line) for the first 5 s. Traces are nearly indistinguishable, confirming functional correctness.

References

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